

ACADEMIC DETAILS				
Degree	Specialization	Institute	Year	CPI/%
B.Tech.	Mechanical Engineering	IIT Gandhinagar	2023-Present	8.39
Class XII	Physics, Chemistry, Maths	Shree Ganesh Junior College, Korhale	2022-2023	81
Class X		Shree Chhatrapati Shivaji Vidyalaya, Kolpe-wadi	2020-2021	85.60

INTERNSHIPS

• Research Intern, SRIP, IIT Gandhinagar

(Advisor - prof. Vinod Narayanan) | [Report Link](#)

Worked on the topic “Shock Wave Boundary Layer Interaction over a Flat Plate with Protrusion”, focusing on heat transfer analysis in supersonic flow (Mach 4.5).

Studied the correlation between wall heat transfer rate and the height/width of a protrusion at a fixed distance from the leading edge.

Learned mesh generation using ANSYS ICEM CFD and improved proficiency in ANSYS Fluent by running 32 RANS-based simulations with the SST k- ω turbulence model.

Developed a Python interpolation model to predict heat transfer for protrusion heights from 0.07 to 0.1 m and widths from 0.025 to 0.2 m.

Validated results with data from a research paper and ensured mesh independence for reliable and accurate simulations.

[May '25 - June '25]

PROJECTS

• Powered Lift Wing Design for STOL Aircraft – LAT Aerospace

(Inter IIT Tech Meet) | [Mid-Term Report](#) | [End-Term Report](#)

Conducted a comprehensive review of **passive** (slats, flaps, high-lift airfoils) and **active** (EBF, USB, circulation control) lift-augmentation techniques, leading to the selection of an **Externally Blown Flap (EBF)** architecture with **Distributed Electric Propulsion (DEP)**.

Evaluated multiple candidate airfoils (E216, E423, NACA 4415, S1223) using XFLR5 and selected the **Selig S1223** for its superior high-lift performance at low Reynolds numbers; subsequently modelled blown-wing aerodynamics using **Kuhn’s semi-empirical method** and a custom inverse-design approach to achieve a target $C_L = 8$, yielding a **wing chord of 0.096 m** and **aspect ratio of 10**.

Developed and executed a complete CFD workflow, modelling propellers as **actuator-disk momentum sources** derived from APC data and Rankine–Froude momentum theory; ensured numerical accuracy through a **mesh-independence study** (lift variation <0.5%) and performed **15 RANS simulations** to study the effects of **propeller placement, flap deflection, and angle of attack**.

Demonstrated **complete stall suppression** up to 45° AoA with a 17.5° flap deflection, achieving $C_L = 6.911$ and **L/D = 5.8**, validating the capability to **lift 15 kg at 20 m/s** for STOL operation; additionally conducted cruise-performance analysis at 80 m/s for clean and flapped configurations to assess aerodynamic efficiency and trim behavior.

[October '25 – December '25]

• Yield Prediction Estimation in Aircraft Wing Spar

(Prof. Harmeet Singh, IIT Gandhinagar) | [Course project \(team of 5\)](#) | [Project Link](#)

Analyzed stress distribution in the I-beam structure of an aircraft wing spar using von Mises and Tresca yield criteria.

Derived theoretical expressions for shear force, bending moment, second moment of area, and calculated principal stresses using Mohr’s Circle.

Implemented these theoretical functions in the backend of a Streamlit web application that takes user inputs for material properties, load distribution, and wing dimensions, and generates shear force, bending moment, and stress distribution plots.

Visualized how the I-beam starts to yield under different load intensities using von Mises and Tresca stress graphs based on the theoretical formulas.

[March '25- April '25]

• Ball-and-Beam Control System: System Identification and Lead Compensator Design

(Prof. Vineet Vashishta , IIT Gandhinagar) | [Course project \(team of 4\)](#) | [Project Link](#)

Performed experimental system identification using chirp (swept-sine) excitation and MATLAB N4SID to obtain a validated third-order continuous-time plant model.

Conducted frequency-domain analysis using Bode, Root Locus, and Nyquist plots, identifying non-minimum phase behavior due to Right Half Plane (RHP) zeros.

Designed a multi-stage cascaded lead compensator to address $\sim 193^\circ$ phase deficiency, reshape dominant pole locations, and improve closed-loop bandwidth.

Verified controller performance through simulation and physical hardware implementation, demonstrating faster transient response compared to PID control.

[October '25 – November '25]

- **Drag Measurement For External Flow Over Objects** [March '25- April '25]
(Prof. Dilip Srinivas Sundaram | Prof. Uddipta Ghosh, IIT Gandhinagar) | Course project (team of 5) | [Project Link](#)
 - Conducted experimental analysis to measure drag forces on a cylinder, NACA0012, and NACA0024 airfoils using a subsonic wind tunnel.
 - Calibrated and recorded velocity profiles at inlet and outlet using a digital micromanometer under the assumption of 2D steady flow.
 - Calculated drag using Reynolds Transport Theorem and velocity profile data; analyzed deviations from theoretical expectations due to experimental setup constraints.
 - Gained hands-on experience in aerodynamic testing and instrumentation techniques.
- **Optimization & ML Algorithms – Gradient Descent, Image Reconstruction, Logistic Regression** [October '25]
(Prof. Nipun Batra, IIT Gandhinagar) | Course project (team of 3) | [Project Link](#)
 - Implemented full-batch and stochastic gradient descent for linear regression on multiple datasets, analyzed convergence to an -neighborhood minimizer, and visualized optimization pathways using contour plots and loss-versus-epoch curves.
 - Applied low-rank matrix factorization for image reconstruction using Gradient Descent and Alternating Least Squares (ALS), recovering missing 30×30 blocks and random pixel regions while evaluating reconstruction quality using RMSE and PSNR metrics.
 - Explored image compression through low-rank approximations of 50×50 patches with varying color complexity, experimented with multiple rank values (5, 10, 25, 50), and compared reconstruction fidelity across compression levels.
- **Decision Tree Implementation from Scratch and Human Activity Recognition** [August '25- September '25]
(Prof. Nipun Batra, IIT Gandhinagar) | Course project (team of 3) | [Project Link](#)
 - Built a decision tree algorithm entirely from scratch, capable of both classification and regression with support for continuous and discrete features. Implemented split criteria including Entropy, Gini Index, and Mean Squared Error.
 - Tested the model on synthetic datasets, automotive fuel-efficiency data, and the UCI Human Activity Recognition (HAR) dataset. Evaluated performance with accuracy, precision, recall, confusion matrices, and cross-validation to determine optimal tree depth.
 - Analyzed runtime against theoretical complexity and benchmarked results against Scikit-learn's DecisionTree implementation. Incorporated PCA and TSFEL-derived features to enhance activity visualization and classification.
- **Metro Journey Planner** [July '25- August '25]
Personal Project | [Project Link](#)
 - Designed and developed a metro route optimization system capable of computing both time-efficient and distance efficient journeys using Dijkstra's Algorithm on a graph generated from CSV datasets.
 - Gained proficiency in graph data structures, and visualization techniques while building an interactive Streamlit web application.
 - Implemented backend logic to detect interchanges, calculate travel time/distance, and render routes with color-coded metro lines for improved user clarity.
 - Deployed the application online, enabling real-time user input for start/end stations and instant display of optimized routes with detailed metrics and interchange information.
- **CFD Analysis of Flow Over a NACA 0012 Airfoil** [April '25]
(Prof. Dilip Srinivas Sundaram | Prof. Uddipta Ghosh, IIT Gandhinagar) | Course project | [Project Link](#)
 - Simulated laminar flow over a NACA 0012 airfoil at 0° and 5° angles of attack using CFD.
 - Calculated lift and drag coefficients to study the effect of angle of attack on lift and drag force.
 - Visualized velocity, pressure, and streamline contours, and post-processed results to understand flow characteristics.
 - Set boundary conditions, created mesh, and improved understanding of symmetric airfoil behavior under varying flow conditions.

TECHNICAL SKILLS

- **Programming Languages:** C, C++, Python, HTML, CSS, JavaScript, SQL.
- **Tools:** MATLAB, MSC Adams, Ansys Fluent, Ansys ICEM CFD, AutoDesk Inventor, AutoDesk Fusion, Mathematica.
- **Libraries:** Numpy, Matplotlib, Pandas, Scikit-learn, Networkx, PyTorch, Cantera.

ACHIEVEMENTS

- Received Academic Citation for achieving academic excellence in Semester 2 (2024–2025) at IIT Gandhinagar.
- Qualified Round 1 of ABU Robocon 2025 with a score of 95/100 as Vice-Captain of IIT Gandhinagar team.

POSITIONS OF RESPONSIBILITY

- **Co-founder and Team vice-captain, Team RUDRA (The ABU Robocon Team of IITGN), IITGN** [October '24-Present]
 - Founded the ABU Robocon team at IIT Gandhinagar and leading its first participation in ABU Robocon 2025. Our team qualified the first round with a score of 95 out of 100.
- **Core Team Member, Mean Mechanics (The Robotics Club at IITGN), IITGN** [August '24-Present]
 - Initiated and coordinated projects as a core member of the Robotics Club at IIT Gandhinagar